

LAB REPORT

Lab Title	STL map container	Lab No.	15
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Lab description/objectives:

1. Demonstrate understanding of the STL map container and its function in cpp.
2. Implement the steps of fruit price management as defined in the lab tasks/requirements.
3. Analyze critical functions and performance about map container.

Source code:

```
#include <iostream>
#include <map>
using namespace std;
void Insertingfruit(map<string, double>& fruitPrices){
    string tmpname;
    double tmpprices;
    cout << "Enter fruit name: ";
    cin >> tmpname;
    cout << "Enter price: ";
    cin >> tmpprices;
    fruitPrices.insert({tmpname, tmpprices});
}
void Printingfruit(map<string, double>& fruitPrices){
    for (auto i = fruitPrices.begin(); i != fruitPrices.end(); i++){
        cout << i->first
            << " : "
            << i->second
            << endl;
    }
}
void Searchingfruit(map<string, double>& fruitPrices){
    string name;
    cout << "Please enter the name of the fruit you want to search."
        << endl;
    cin >> name;
    auto tmp = fruitPrices.find(name);
    if (tmp != fruitPrices.end()){
        cout << "The price of "
            << name
```

```
<< " is "
<< tmp->second
<< "."
<< endl;
} else {
    cout << "The fruit is not in the list."
    << endl;
}
}
void ModifyingPrice(map<string, double>& fruitPrices){
    string name;
    double newprice;
    cout << "Please enter the fruit you want to modify the price and its new price."
        << endl;
    cin >> name >> newprice;
    auto tmp = fruitPrices.find(name);
    if (tmp != fruitPrices.end()){
        tmp->second = newprice;
        cout << "The price has been successfully modified."
            << endl;
    } else {
        cout << "Error. The fruit cannot be found for price modification."
            << endl;
    }
}
void Deletingfruit(map<string, double>& fruitPrices){
    string name;
    cout << "Please enter the fruit you want to delete."
        << endl;
    cin >> name;
    auto tmp = fruitPrices.find(name);
    if (tmp != fruitPrices.end()){
        fruitPrices.erase(tmp);
        cout << "The fruit has been successfully deleted."
            << endl;
    } else {
        cout << "Error. The fruit cannot be found for deletion."
            << endl;
    }
}
int main(){
    map<string, double> fruitPrices;
    int num;
```

```
cout << "Please enter the number of initial elements in the fruit list."
    << endl;
cin >> num;
for (int i = 0; i < num; i++){
    Insertingfruit(fruitPrices);
}
cout << "The fruit list is:"
    << endl;
Printingfruit(fruitPrices);
Searchingfruit(fruitPrices);
ModifyingPrice(fruitPrices);
Deletingfruit(fruitPrices);
cout << "The modified fruit list is:"
    << endl;
Printingfruit(fruitPrices);
return 0;
}
```

Program output:

```
Please enter the number of initial elements in the fruit list.
3
Enter fruit name: apple
Enter price: 8
Enter fruit name: banana
Enter price: 3
Enter fruit name: orange
Enter price: 4
The fruit list is:
apple : 8
banana : 3
orange : 4
Please enter the name of the fruit you want to search.
banana
The price of banana is 3.
Please enter the fruit you want to modify the price and its new price.
apple 5
The price has been successfully modified.
Please enter the fruit you want to delete.
orange
The fruit has been successfully deleted.
The modified fruit list is:
apple : 5
banana : 3
```

```
Please enter the number of initial elements in the fruit list.
3
Enter fruit name: apple
Enter price: 5
Enter fruit name: banana
Enter price: 3
Enter fruit name: orange
Enter price: 4
The fruit list is:
apple : 5
banana : 3
orange : 4
Please enter the name of the fruit you want to search.
watermelon
The fruit is not in the list.
Please enter the fruit you want to modify the price and its new price.
bear
4
Error. The fruit cannot be found for price modification.
Please enter the fruit you want to delete.
bear
Error. The fruit cannot be found for deletion.
The modified fruit list is:
apple : 5
banana : 3
orange : 4
```

Analysis:

1. Key Challenges & Solutions

- Understand the various functions in the map container and their corresponding usage.
- Use appropriate iterator to iterate through the container and use the type *auto* to perform.

2. Result Validation

- Confirm passing reference to the function so that the map in the main function can be modified correctly.
- Print the error message when the fruit is not in the list.

3. Improvement Suggestions

- Can check the data type when entering the number of initial elements.

Conclusion:

In cpp, the map container can be used to correspond key to the value and provides various function to insert, delete, modify, search in the container.